Group 1 Execution Plan

PPO $\times$ Macro Adaptive Strategy

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March 30, 2026

Outline

1. Strategy Recap
2. Execution Design
3. Order Types & Execution
4. Transaction Costs
5. Trade Lifecycle & Record Keeping
6. Backtest $\rightarrow$ Live

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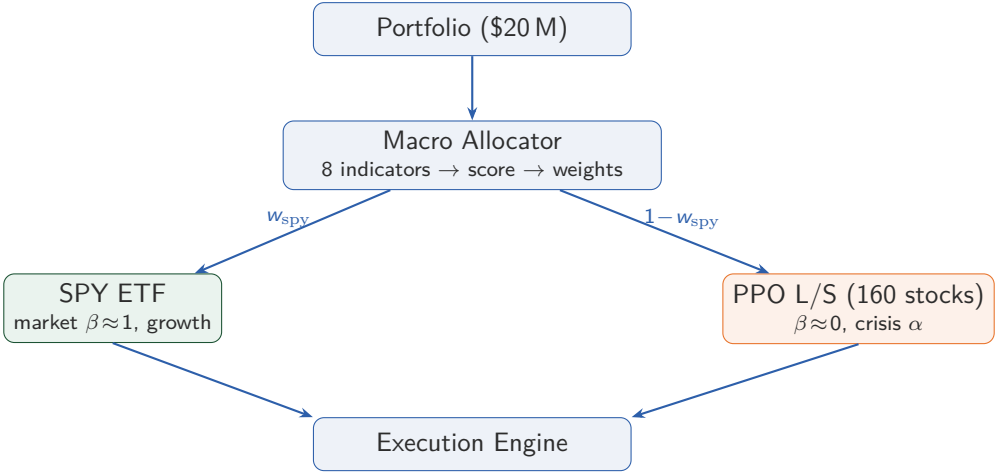
Core Insight: Crisis Alpha from Beta Neutrality

- A **PPO reinforcement-learning agent** dynamically tunes 15 factor weights to build a **beta-neutral long/short equity portfolio** ($\beta \approx 0$, 80 L / 80 S from top-400 US equities).
- Beta neutrality $\Rightarrow$ the portfolio is uncorrelated with the market $\Rightarrow$ significant **crisis alpha** during sell-offs.
- **Trade-off:** zero beta means limited upside in bull markets.
- **Solution:** combine PPO with a **SPY position** ($\beta \approx 1$) and allocate capital dynamically via a macro allocator.
  - Bull regime $\rightarrow$ more SPY $\rightarrow$ ride market growth.
  - Crisis regime $\rightarrow$ more PPO $\rightarrow$ harvest crisis alpha.

Two-Layer Architecture



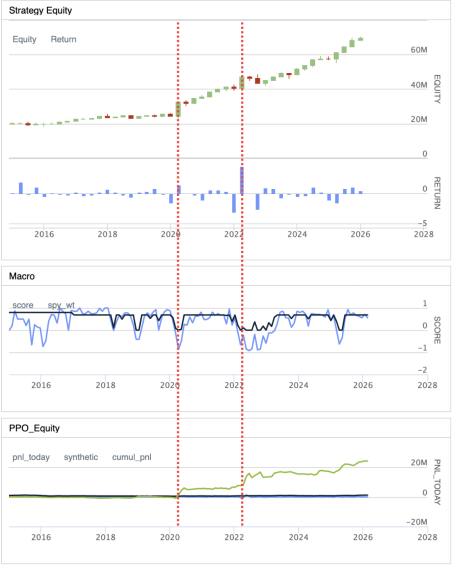
The macro allocator's sensitivity is critical: too sensitive → excess turnover and false signals; too slow → miss regime shifts.

Ramp-Up Period: Protecting PPO During Training

- PPO begins with a **random policy**—early trades are exploratory and loss-making.
- Without protection, large random L/S positions would trigger **margin calls**, killing execution and preventing learning.
- **Our design:** for the first 2 years, fix allocation to **80 % SPY** / **20 % gross PPO**.
  - SPY growth keeps the portfolio solvent and covers PPO losses.
  - PPO explores freely within a small sandbox without margin risk.
- After ramp-up, the macro allocator takes over with full dynamic allocation.

	Ramp-Up (2 yr)	**Post Ramp-Up**
SPY weight	80% (fixed)	dynamic
PPO gross	20% (fixed)	dynamic
PPO policy	random	trained

Performance Across Regimes



Backtest Performance vs. Benchmark

Metric	Our Strategy	SPX
Annual Return	11.48 %	**11.54 %**
Annual SD	**9.70 %**	14.86 %
Sharpe (rf = 0)	**1.18**	0.78
Calmar (rf = 0)	**0.83**	0.34
Max Drawdown	**13.80 %**	33.93 %
Beta	**0.376**	1
Alpha	**0.030**	0

Backtest: Jan 2015 – Mar 2026, $20 M start capital. Transaction cost = IB tiered commissions.

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Execution Objectives

- **Trade at identical times as backtest.**
  - Daily rebalance 15 min after open via `Schedule.On`; signal and execution in the same scheduled event.
- **Trade at backtest prices or better.**
  - Three modes available: passthrough / limit-at-mid / MOC.
- **Minimise slippage.**
  - 0.5 bps one-way slippage model; limit orders target mid-price; less-liquid assets traded first.
- **Minimise commission.**
  - IB tiered pricing: $0.0035/sh, $0.35 min; identical in backtest and live.
- **Minimise tax.**
  - 30 % dividend WHT tracked per event; penalised in PPO reward.

Executability: When Do We Trade?

- U.S. closing prices are determined by a **closing auction**—signals generated at close cannot execute at that same close price.
- We generate signals **15 min after market open** each day. The previous-day close is already fully known.
  - For our daily holding period, the open-vs-close gap is negligible.
- Execution follows **immediately** within the same scheduled event—no gap between signal and order.
  - Eliminates the uncertainty of trading at the next open (QuantConnect default).
- **No after-hours trading.** Bid-ask spreads widen and liquidity drops.
- **MOC fallback** available: `MarketOnCloseOrder` before the 3:45 pm cutoff.

Instrument Selection: What Do We Trade?

- Priority: Futures > Stocks/ETFs > Options.
- **Macro layer: SPY ETF.**
  - We chose SPY over ES futures: no rolling risk, no spread-order complexity, simpler bookkeeping for daily rebalancing.
  - Dividend WHT is indeed a problem.
- **PPO layer: 160 individual US equities** (80 L / 80 S).
  - Universe: top 400 by dollar volume, price > \$5, with fundamental data.
  - Individual stocks are required because the PPO agent needs factor-level tilts—ETFs cannot provide this granularity.

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Three Execution Modes

- **Passthrough** (`SetHoldings`).
  - Used in backtest and low-urgency scenarios. Ensures zero code divergence between backtest and live.
- **Limit at mid-price** (default in live). 90 s timeout $\rightarrow$ market fallback.
  - Targets the mid-price for price improvement. The timeout guarantees the full rebalance completes each day, even if liquidity is thin.
- **Market-on-Close** (`MarketOnCloseOrder`).
  - For close-price fidelity. Orders submitted before 3:45 pm cutoff. Useful if we later switch signals to close-price based.

All three modes share the **same cost models** (slippage + commission), so backtest fills $\approx$ live fills.

Smart Order Routing

- **Less-liquid-first ordering.** Orders are sorted by descending bid-ask spread; hardest fills go first via limit, easier fills follow.
  - Pairs-trading: trade the less liquid leg first. Applied here across all 160 stocks.
- **Chunked execution** for large orders ($> \$500\,\mathrm{K}$): split into 4 child orders to reduce market impact.
  - Averages out timing uncertainty.
- **Timeout fallback.** Unfilled limits after 90 s are cancelled and resubmitted as market orders.
  - Guarantees complete daily rebalance—critical for maintaining the target beta-neutral portfolio.

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Slippage Model

- We use a **fixed-percentage one-way slippage** of 0.5 bps (1 bps round-trip), implemented as `SpreadSlippageModel`.
  - Models half the bid-ask crossing cost. Conservative for liquid US large-caps in our top-400 universe.
- Applied to **every equity** via `OnSecuritiesChanged`—no asset is unmodelled.
- The same slippage model runs in both backtest and live.
- Our execution target is the **mid-price**. VWAP is a viable upgrade if order sizes grow.

$$\text{Slippage per side} = P \times 0.00005 \quad (0.5 \text{ bps})$$

Commission Model

- We use **IB tiered pricing**, implemented as `IBTieredFeeModel`.
  - \$0.0035/share, \$0.35 minimum per order, capped at 0.5 % of trade value.
  - Cheaper than IB fixed (\$0.005/sh, \$1 min)—keeps backtest impact realistic but not overly conservative.
- The same fee model is used in backtest and live.

$$\text{Fee} = \max\big(\$0.35,\ \min(Q \times \$0.0035,\ Q \cdot P \cdot 0.005)\big)$$

Short Selling & Borrow Costs

- PPO holds $\sim$80 short positions at any time ($\sim$47.5 % of PPO capital).
- **Borrow cost:** 3 % p.a. flat rate in live, accrued daily.
- **Our mitigations:**
  - Liquid universe (top-400 by $ vol) $\rightarrow$ mostly "general collateral", avoiding special-situation names.
  - `RiskGuard` warns if any bid-ask spread $>$ 200 bps.
  - `check_short_feasibility` flags symbols **before** the order is sent.
  - Daily short-inventory summary logged for monitoring.

$$\text{Daily borrow charge} = \sum_{i:\, h_i < 0} |h_i \cdot P_i| \times \frac{0.03}{252}$$

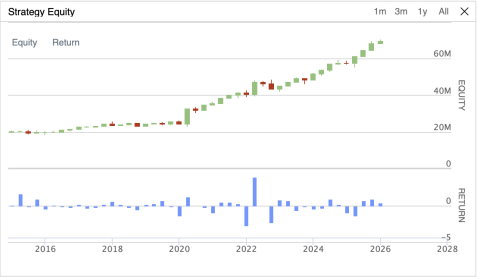
Dividend Withholding Tax

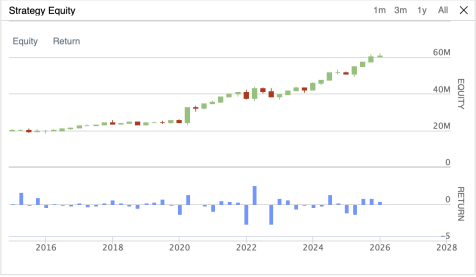
- Non-US residents face **30 % US withholding tax** on equity dividends—a material drag for a 160-stock portfolio.
- `DividendTaxModel` scans every data slice for dividend events on **long holdings only**.
  - Short positions pay dividends to the lender (broker-handled, no WHT).
- Tax is injected into the **PPO reward function** ($-\lambda_{\text{tax}} \cdot \text{tax}\,\%$), so the agent *learns* to avoid high-dividend longs.
  - Dividends can incur large tax cost for stocks/ETFs.
- Full audit trail persisted to `ObjectStore`: symbol, qty, gross dividend, tax, cumulative total.

$$\text{WHT} = \text{Distribution}_{\text{per share}} \times \text{Qty} \times 30\,\%$$

Impact of Real-World Frictions on Performance

- The following comparison shows strategy equity curves **without** and **with** full friction costs.





[Left: no friction Right: with friction]

Friction	Slippage	Commission	Borrow	Div WHT
Model	0.5 bps/side	\$0.0035/sh	3 % p.a.	30 %

Impact of Real-World Frictions on Performance (continued)

Metric	SPX	Pre-Friction	Post-Friction
Annual Return	**11.54 %**	11.48 %	10.17 %
Annual SD	14.86 %	**9.70 %**	**9.70 %**
Sharpe (rf = 0)	0.78	**1.18**	1.05
Calmar (rf = 0)	0.34	**0.83**	0.63
Max Drawdown	33.93 %	**13.80 %**	16.20 %
Beta	1	**0.376**	0.370
Alpha	0	**0.030**	0.022
Avg Turnover	–	**27.74 %**	27.78 %
Win Rate	–	**50 %**	**50 %**

Pre-Friction = IB tiered commissions only. Post-Friction = + 0.5 bps slippage + 3 % p.a. borrow + 30 % dividend WHT.

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Trade Signal Types

- **Entry Signal:** open a new long or short position.
  - Triggered when PPO factor weights produce scores that rank a stock into the top-80 (long) or bottom-80 (short).
- **Exit Signal:** close all positions in a trade.
  - Triggered by crisis override ($g_{\text{ppo}} \to 0$, full capital to SPY) or when a stock leaves the universe.
- **Rebalance Signal:** adjust position size without full exit.
  - Triggered daily by macro re-weighting, beta adjustment, volatility targeting, or portfolio-level investment/redemption.
- Each signal creates, updates, or closes a **Trade Order** tracked by `TradeTracker` with unique IDs—preserving the full P&L audit trail (signal $\to$ trade-order flow).

Five-Layer Record Keeping

- **Layer 1 — Signal snapshot.** Data and market state at signal time; compare live vs. backtest to detect divergence.
- **Layer 2 — Batch signals.** All 160 PPO target values + macro metadata per rebalance.
- **Layer 3 — Fill execution.** Actual fill price, qty, commission; slippage vs. mid computed per fill.
- **Layer 4 — Daily MTM.** Unrealised P&L per holding at close; for daily return, volatility, drawdown series.
- **Layer 5 — Realised P&L.** `TradeTracker` records profit per trade ID including dividends and commissions.

All five layers run in **both backtest and live** and persist to QC `ObjectStore`—enabling full post-trade analysis and broker reconciliation.

Pre-Trade Risk Guards

- **Circuit breaker:** halt all trading if daily loss $> 10\,\%$.
  - With 160 positions, a correlated move against us can be rapid. The circuit breaker prevents compounding losses.
- **Gross leverage cap:** block rebalance if gross leverage $> 1.5\times$.
  - Our target is $0.95\times$ PPO + SPY weight. Exceeding $1.5\times$ signals an execution error.
- **Single position cap:** reject if any holding $> 20\,\%$ of equity.
  - Prevents concentration risk in any one name.
- **Spread alert:** warn (non-blocking) if bid-ask $> 200\,\text{bps}$.
  - Flags potential liquidity problems before the order is sent.
- E-mail notification on circuit-breaker trigger.

End-of-Day Reconciliation

- **Broker vs. target check:** flag any symbol deviating $> 2\,\%$.
  - With 160 positions, small fill errors can accumulate. Daily reconciliation catches drift before it compounds.
- **Cross-check** broker positions against `TradeTracker` open positions.
  - Detects qty mismatches and unexpected holdings not in our target set.
- **Dividend income** synced into `TradeTracker` in live mode.
  - Ensures realised P&L includes all cash flows, not just price changes.
- **Auto-correction** available but **disabled by default**.
  - We prefer to flag and review manually to avoid corrective trades amplifying errors.
- Execution stats (avg/max slippage, total commissions) logged daily.

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Model Registry & Seamless Transition

- **Model Registry** saves full PPO state to `ObjectStore`: policy weights, value weights, *and both Adam optimizer states*.
  - Preserving optimizer momentum means live training continues *identically* from where backtest left off.
- Backtest saves a checkpoint every 20 days, plus a final `backtest_final` checkpoint tagged with Sharpe, cumulative P&L, and max drawdown.
- **Live** loads the best checkpoint by Sharpe.
- Fallback chain: Registry → ObjectStore (legacy) → filesystem → random weights (with alert).

Design Principle

Backtest *is* the first phase of live trading. Every cost model, risk guard, and record layer is active in **both** modes—zero code divergence.

Execution Plan Summary

Dimension	Decision
Automated or manual?	Fully automated (QuantConnect $\rightarrow$ IB paper/live)
Data source	QC equity + CBOE VIX + FRED yield curve
Signal generation	15 min after market open (daily)
Execution time	Immediately after signal; MOC available
SPY instrument	SPY ETF (no rolling, simpler than ES futures)
PPO instruments	Top-400 liquid US equities; long 80 / short 80
Order type	Limit at mid, 90 s timeout $\rightarrow$ market
Slippage model	0.5 bps one-way (`SpreadSlippageModel`)
Commission	IB tiered \$0.0035/sh (`IBTieredFeeModel`)
Borrow cost	3 % p.a. (live); daily accrual on shorts
Dividend tax	30 % WHT on long dividends
Risk controls	Circuit breaker -10 %; leverage $\leq 1.5\times$; position ≤ 20 %
Record keeping	5 layers (signal / batch / fill / MTM / realised)

Thank You

Questions & Discussion

Appendix A1: PPO Network Architecture

Policy Network $\pi_\theta(a \mid s)$: 3-layer MLP (128 hidden units each, Tanh activations). Outputs mean $\mu(s)$ and learned log-std $\log \sigma$.

$$a \sim \mathcal{N}\big(\mu_\theta(s),\ \mathrm{diag}(\sigma^2)\big), \qquad w = \frac{\tanh(a)}{\|\tanh(a)\|_1}$$

$w \in \mathbb{R}^{15}$ are L1-normalised factor weights.

Value Network $V_\phi(s)$: 3-layer MLP (128 hidden, Tanh) $\rightarrow$ scalar value estimate.

State vector ($d = 87$): 12 core features + 5×15 factor statistics.
Core: log(equity), cash%, gross lev, β, w_L, w_S, n_{pos}, turnover, tc%, borrow%, β_{last}, TE.
Per factor ($k = 1 \ldots 15$): $\overline{\text{IC}}_k$, $\sigma_{\text{IC},k}$, $t_{\text{IC},k}$, $\overline{\text{QS}}_k$, $\sigma_{\text{QS},k}$.

Action ($d = 15$): one weight per alpha factor.

Appendix A2: PPO Objective & GAE

Clipped Surrogate Objective:

$$L^{\text{CLIP}}(\theta) = \hat{\mathbb{E}}_t\Big[\min\big(r_t(\theta)\,\hat{A}_t,\ \ \text{clip}(r_t(\theta),\,1-\epsilon,\,1+\epsilon)\,\hat{A}_t\big)\Big], \quad r_t(\theta) = \frac{\pi_\theta(a_t \mid s_t)}{\pi_{\theta_{\text{old}}}(a_t \mid s_t)},\ \epsilon = 0.2$$

Generalised Advantage Estimation ($\gamma = 0.99,\ \lambda = 0.95$):

$$\hat{A}_t = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l\,\delta_{t+l}, \qquad \delta_t = r_t + \gamma\,V(s_{t+1}) - V(s_t)$$

Value loss: $L^V(\phi) = \frac{1}{N}\sum_t (V_\phi(s_t) - \hat{R}_t)^2, \quad \hat{R}_t = \hat{A}_t + V(s_t).$

Training: 10 epochs, batch size 256, gradient clip 1.0. Updated every 5 trading days (after ramp-up ends).

Appendix A3: Macro Score — 8 Indicators

The macro allocator computes a weighted vote:

$$s_{\text{macro}} = \frac{\sum_i v_i \cdot \omega_i}{\sum_i \omega_i}, \qquad v_i \in \{-1,\, 0,\, +1\}$$

i	**Indicator**	ω_i	**Bullish** $(+1)$	**Bearish** (-1)
1	SPY / SMA200 − 1	2.0	$> 3\%$	$< -2\%$
2	SMA50 / SMA200 − 1	1.5	$> 1\%$	$< -1\%$
3	252-d momentum	1.0	$> 5\%$	$< -5\%$
4	Drawdown from 252-d hi	2.0	$> -3\%$	$< -8\%$
5	VIX level	2.5	< 16	> 22
6	Yield 10Y−2Y	1.0	> 0.50	< 0
7	20-d momentum	2.0	$> 3\%$	$< -4\%$
8	50-d momentum	1.5	$> 4\%$	$< -5\%$

Indicators between both thresholds vote 0 (neutral). Total weight $\Sigma\omega = 13.5$; score range $\approx [-1, +1]$.

Appendix A4: Score Dynamics — Trend, Vol, Acceleration

From the score buffer (up to 320 observations):

Trend (fast mean − slow mean):

$$\text{trend} = \bar{s}_5 - \bar{s}_{20}$$

Volatility (rolling std of recent scores):

$$\text{vol} = \text{std}(s_{-10:})$$

Volatility acceleration (change in short-term vol):

$$\text{vol_accel} = \text{std}(s_{-5:}) - \text{std}(s_{-10:-5})$$

These three quantities feed directly into the crisis intensity computation. They capture the **speed and direction** of regime changes, not just the current score level. A sharp negative trend combined with rising volatility signals a crisis is **accelerating**.

Appendix A5: Crisis Intensity — Four Signals

Four normalised crisis signals (all clamped to $[0, 1]$):

$$\sigma_{\text{trend}} = \text{clamp}\left(\frac{-\text{trend} - 0.03}{0.22},\ 0,\ 1\right), \qquad \sigma_{\text{vol}} = \text{clamp}\left(\frac{\text{vol} - 0.05}{0.20},\ 0,\ 1\right),$$

$$\sigma_{\text{accel}} = \text{clamp}\left(\frac{\text{vol_accel} - 0.01}{0.09},\ 0,\ 1\right), \qquad \sigma_{\text{level}} = \text{clamp}\left(\frac{-s_{\text{macro}} - 0.10}{0.30},\ 0,\ 1\right).$$

Composite raw crisis:

$$c_{\text{raw}} = 0.50\ \min(\sigma_{\text{trend}},\ \sigma_{\text{vol}}) + 0.25\ \sigma_{\text{vol}} \cdot \sigma_{\text{accel}} + 0.25\ \sigma_{\text{level}}$$

The primary term requires **both** negative trend and high volatility simultaneously (the minimum operator). The bonus terms reward acceleration and deeply negative macro scores.

Appendix A6: Crisis Intensity — Asymmetric Smoothing

The raw crisis score is smoothed with an **asymmetric EMA**—fast ramp-up, slow decay:

$$c_t = \alpha\, c_{\text{raw}} + (1-\alpha)\, c_{t-1}$$

$$\alpha = \begin{cases} 0.70 & c_{\text{raw}} > c_{t-1} \ \text{(entering crisis — react fast)} \\ 0.60 & \text{trend} > 0.05 \ \wedge \ s > -0.5 \ \text{(strong recovery signal)} \\ 0.40 & \text{trend} > 0 \ \text{(mild recovery)} \\ 0.15 & \text{otherwise (slow decay — stay defensive)} \end{cases}$$

Why asymmetric?

- Entering crisis: $\alpha = 0.70$ means 70 % of the new signal is adopted immediately. We shift capital to PPO within 1–2 days.
- Exiting crisis: $\alpha = 0.15$ means we wait $\sim$6–7 days to confirm recovery before shifting back to SPY. Avoids whipsaw from dead-cat bounces.

Appendix A7: SPY Weight — C-Style Dynamic Map

We use only the C-style map (`dynamic_w` = 1, `static_w` = 0):

$$w_c(s) = \text{clamp}\Big(\underbrace{0.7}_{\text{base}} + \frac{s}{\hat{\sigma}_{\text{long}}},\ 0,\ 1\Big), \qquad \hat{\sigma}_{\text{long}} = \text{std}\big(s_{\text{buf}}[-260:]\big)$$

Intuition: when the macro score is positive and large relative to its historical variability, SPY gets high weight. When the score is negative, SPY weight drops toward zero—capital flows to PPO.

Crisis-adjusted SPY target:

$$w_{\text{spy}} = w_c \times (1 - c_t)$$

Floor constraint: PPO always gets at least 30 % gross.

$$w_{\text{spy}} \ \leq\ 1 - \frac{g_{\min}}{0.95} \ =\ 1 - \frac{0.30}{0.95} \ \approx\ 0.684$$

Appendix A8: PPO Gross Exposure and Capital

Given the SPY weight, the remaining capital goes to PPO:

PPO gross exposure:

$$g_{\text{ppo}} = 0.95\,(1 - w_{\text{spy}})$$

PPO capital:

$$K_{\text{ppo}} = (1 - w_{\text{spy}}) \times \text{Equity}$$

Worked example:

$$s_{\text{macro}} = 0.4, \quad c_t = 0.1, \quad \hat{\sigma}_{\text{long}} = 0.25, \quad \text{Equity} = \$20\text{M}$$

Step	Computation
C-style base	$w_c = \text{clamp}(0.7 + 0.4/0.25,\ 0,\ 1) = \text{clamp}(2.3) = 1.0$
Crisis adjust	$w_{\text{spy}} = 1.0 \times (1 - 0.1) = 0.90$
Cap enforce	$w_{\text{spy}} = \min(0.90,\ 0.684) = 0.684$
PPO gross	$g_{\text{ppo}} = 0.95 \times (1 - 0.684) = 0.300$
PPO capital	$K_{\text{ppo}} = 0.316 \times \$20\text{M} = \$6.32\text{M}$

Appendix A9: Full Allocation Pipeline

8 Macro Indicators → Weighted Vote → Score Buffer → Trend, Vol, Vol Accel → Crisis Intensity c_t → w_{spy}, g_{ppo}

Summary of key formulas:

$$s_{\text{macro}} = \frac{\sum v_i \omega_i}{\sum \omega_i} \qquad \text{(weighted vote of 8 indicators)}$$

$$c_t = \alpha\, c_{\text{raw}} + (1-\alpha)\, c_{t-1} \qquad \text{(asymmetric EMA, } \alpha \in \{0.15, 0.40, 0.60, 0.70\}\text{)}$$

$$w_{\text{spy}} = \text{clamp}\big(0.7 + s/\hat{\sigma},\ 0,\ 1\big) \times (1 - c_t) \qquad \text{(C-style} \times \text{crisis discount)}$$

$$g_{\text{ppo}} = 0.95\,(1 - w_{\text{spy}}) \qquad \text{(residual to PPO)}$$

Appendix A10: Ramp-Up Period Allocation

During the first 2 years (2015-01-01 to 2017-01-01), allocation is **fixed** regardless of macro conditions:

$$w_{\text{spy}} = 0.80, \qquad g_{\text{ppo}} = 0.20$$

PPO capital during ramp-up:

$$K_{\text{ppo}} = \text{Equity} \times \frac{g_{\text{ppo}}}{\text{gross_budget_frac}} = \text{Equity} \times \frac{0.20}{0.95} \approx 0.211 \times \text{Equity}$$

Why this design?

- PPO's random policy creates unpredictable L/S positions.
- 80 % SPY provides a stable capital base, preventing margin calls.
- Even a total wipeout of the PPO book costs only ∼10 % of portfolio.
- Macro dynamics (trend, vol, crisis) are still tracked internally, so the allocator is "warm" when ramp-up ends.

Appendix A11: 15 Alpha Factors

k	Factor	Formula (sign = desired direction)
1	mom_5	$P_t/P_{t-5} - 1$
2	mom_20	$P_t/P_{t-20} - 1$
3	mom_60	$P_t/P_{t-60} - 1$
4	rev_2	$-(P_t/P_{t-2} - 1)$
5	rev_5	$-(P_t/P_{t-5} - 1)$
6	vol_10	$-\mathrm{std}(r_{t-9:t})$
7	vol_20	$-\mathrm{std}(r_{t-19:t})$
8	hlv_20	$-\mathrm{mean}\big((H-L)/P\big)_{20}$
9	rsi_14	$-\mathrm{RSI}(14)$
10	trend_20	$P_t/\mathrm{MA}_{20} - 1$
11	gap_1	$P_t/P_{t-1} - 1$
12	skew_20	skewness of r_{20}
13	kurt_20	excess kurtosis of r_{20}
14	dvol_20	$-\mathrm{std}(r_{<0})_{20}$
15	maxdd_60	max drawdown over 60 d

All factors are cross-sectionally winsorised (1 %) then z-scored. PPO outputs $w \in \mathbb{R}^{15}$; stock score $= Z_i^\top w$.

Appendix A12: Portfolio Construction & Beta Hedge

Scoring: $\text{score}_i = Z_i^\top w, \quad Z \in \mathbb{R}^{N\times 15}, \quad w \in \mathbb{R}^{15}$ (PPO action).

Selection: Top-80 by score $\to$ long; Bottom-80 $\to$ short.

Weighting: Within each side, weight $\propto |\text{score}|$; rescaled so $\sum |h_L| = \frac{0.95}{2} K_{\text{ppo}}$ (same for short side).

Position bounds: $h_{\min} = 0.5\,\% \cdot K_{\text{ppo}}, \quad h_{\max} = 15\,\% \cdot K_{\text{ppo}}$.

Beta neutralisation (up to 40 iterations):

1. Compute $\hat{\beta} = \sum_i h_i \beta_i / K_{\text{ppo}}$.
2. If $|\hat{\beta}| \leq 0.10$, stop.
3. Find best (i, j) pair (one long, one short) whose $\beta_i - \beta_j$ can tilt $\hat{\beta}$ toward 0.
4. Shift Δ dollars from j to i subject to position bounds.
5. Repeat.

Appendix A13: PPO Reward Function

The agent receives a **PPO-isolated return** (SPY contribution stripped):

$$r_{\text{ppo}} = \frac{r_{\text{portfolio}} - w_{\text{spy}} \cdot r_{\text{SPY}}}{g_{\text{ppo}}}$$

Composite reward:

$$R = r_{\text{ppo}} - \underbrace{\lambda_\beta \max(0, |\beta| - 0.10)}_{\text{beta penalty}} - \underbrace{\lambda_f \cdot \text{fee\%}}_{\text{commission}} - \underbrace{\lambda_b \cdot \text{borrow\%}}_{\text{borrow cost}} - \underbrace{\eta \cdot \text{turnover\%}}_{\text{turnover}} - \underbrace{\lambda_\tau \cdot \text{tax\%}}_{\text{div WHT}}$$

Coefficient	Value
λ_β	2.0
λ_f	1.0
λ_b	1.0
η	0.05
λ_τ	1.0

Appendix A14: Transaction Cost Formulas

Slippage (`SpreadSlippageModel`):

$$\text{slip} = P \times 0.00005 \quad \text{per side } (0.5\,\text{bps})$$

Commission (`IBTieredFeeModel`):

$$\text{fee} = \max\Big(\$0.35,\ \ \min(Q \times \$0.0035,\ \ Q \cdot P \cdot 0.005)\Big)$$

Daily borrow charge:

$$\text{borrow}_t = \sum_{i:\, h_i < 0} |h_i \cdot P_i| \ \times\ \frac{0.03}{252}$$

Dividend withholding tax:

$$\text{WHT}_j = D_j \times Q_j \times 0.30 \qquad (Q_j > 0 \text{ only})$$

Total execution cost per period:

$$C = \sum \text{slippage} + \sum \text{fee} + \sum \text{borrow}_t + \sum \text{WHT}_j$$